

Tristan Lazard

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EDUCATION

PSL University, Paris, 75006
ITI - MS Innovation, Entrepreneurship and innovation. 2018-2019.

Sorbonne University, Paris 2016-2017
Master 2 : Applied mathematics Thesis: "A model of random mutualistic network."

Ecole Normale Supérieure, rue d'Ulm, Paris
Master 1 : Evolutionary biology and ecology 2015-2016
Bachelor in life-sciences 2013-2014

Lycée Saint-Louis, Paris 2011-2013
Classe préparatoire BCPST

AgroCampus-Ouest, Rennes 2014-2015
Bachelor in Agronomy, spare year

SKILLS

Languages: French (mother tongue), fluent English and debutant in Spanish.
Programming languages: Python, Bash, $\text{\LaTeX}$, nextflow.
Computing skills: docker, slurm, cloud-computing, GPU-computing, ML+image processing usual libs (pytorch, openCV, PIL, sklearn...).

PROFESSIONAL EXPERIENCE

PhD in BioInformatics CBIO - Mines-Paristech
December 2019 - December 2023 Paris, France
My PhD project focuses on processing Whole Slide Images, which are digitized microscopic images of diseased tissue commonly used by clinicians for diagnosis. These images contain a vast amount of information, leading to large file sizes (10-15 GB uncompressed). By developing machine learning algorithms to predict patient-level labels from these images, we can assist clinicians in their practice and improve our understanding of the disease through the interpretation of the algorithm's decisions. Main Achievements:

- **5 publications**, 3 first authors, in biological journals and machine learning venues.
- 1 patent.
- **Prize-winning solution (5000\$)** at the VisioMel challenge. Link to the solution

Engineering internship CMM, Mines-Paristech
Jan - July 2019 Fontainebleau, France
Automatic counting of Melanocytes: In partnership with L'Oréal, I created an algorithm for automatically counting melanocytes in microscopic images. This solution has been integrated into the processing pipeline and is now utilized in regular research.

Research internship IBPS, Sorbonne university
Jan - August 2018 Paris, France
Large scale phylogenetic tree inference: Develop an algorithm for calculating phylogenetic trees from huge genetic sequence data (-omics data). .

- PUBLICATIONS** **Lazard, T.**, Bataillon, G., Naylor, P., Popova, T., Bidard, G.-C., Stoppa-Lyonnet, D., Stern, M.- H., Decencie‘ re, E., Walter, T., and Vincent-Sal- omon, A. (2022). Deep Learning identifies new morphological patterns of Homologous Recombination Deficiency in luminal breast cancers from whole slide images. **Cell Reports Medicine**
- Lazard, T.**, Lerousseau, M., Decenci re, E., Walter, T. (2023). Giga-SSL: Self-Supervised Learning for Gigapixel Images. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) **Workshops**.
- Lazard, T.**, Blusseau S., Velasco-Forero S, Decenci re E., Flouret V., Cohen C., Baldewick T. (2022). Applying Deep Learning to Melanocyte Counting on Fluorescent TRP1 Labelled Images of In Vitro Skin Model. **Image Analysis Stereology**.
- Lubrano M., **Lazard T.**, Balezo G., Bellahsen-Harrar Y., Badoual C., Berlemont S., Walter T. (2022). **Computer Vision–ECCV 2022 Workshops**: Tel Aviv, Israel, October 23–27, 2022, Proceedings.
- Naylor P, **Lazard T**, Bataillon G, La  M, Vincent-Salomon A, Hamy AS, et al. Prediction of Treatment Response in Triple Negative Breast Cancer From Whole Slide Images. **Front Signal Process**.
- Lazard T.**, Decenci re E., Walter T. Democratizing computational pathology: optimized WSI representations for the TCGA. **Preprint**.
- T. Lazard**, et. al., Cancer du sein - Utilisation de l’intelligence artificielle pour pr dire le statut tumoral relatif   la recombinaison homologue (2023). **M decine et Sciences**
- A. Beaufr re*, **T. Lazard***, et. al. Self-supervised learning to predict intrahepatic cholangiocarcinoma transcriptomic classes on routine histology (2024). **Preprint**

INTERESTS Climbing, harmonica, poetry and litterature.